SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool: Scenario-Based Task (Medium)
Assessment Tool Weightage: 20%

Q. NO	Question	Bloom's Level
1	A university stores student and course data in a single table. Analyze the structure and identify normalization issues, then redesign it up to 3NF	. BL4 – Analyze
2	A company database contains employee and department details. Design appropriate SQL queries to retrieve employees working in multiple departments and justify your approach	. BL6 – Create
3	Given a relation with multiple functional dependencies, determine the candidate keys and check whether the relation satisfies BCNF. If not, decompose it.	BL5 – Evaluate
4	A banking system requires secure data access. Design views and constraints to ensure data security and integrity for different types of users.	BL6 – Create
5	A sales database contains redundant data leading to anomalies. Analyze the schema and apply normalization up to 3NF or BCNF to eliminate redundancy.	BL4 – Analyze
6	Given a dataset of customers and orders, write SQL queries using joins and subqueries to retrieve meaningful information such as frequent customers.	BL3 – Apply
7	A relation contains multi-valued dependencies. Analyze the structure and decompose it into 4NF, explaining the reasoning.	BL4 – Analyze

8	A database designer used improper constraints, leading to inconsistent data. Evaluate the schema and suggest appropriate integrity constraints.	BL5 – Evaluate
9	For a given relational schema, analyze the presence of join dependencies and decompose it into 5NF to remove redundancy.	BL6 – Create
10	A system uses inefficient SQL queries for data retrieval. Analyze the queries and optimize them using joins, indexing concepts, or query restructuring.	BL5 – Evaluate

Code of Conduct:

I**, A Mohammad Naheed - 192525336** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A Mohammad Naheed

1. A university stores student and course data in a single table. Analyze the structure and identify normalization issues, then redesign it up to 3NF

```
CREATE TABLE Student (
   StudentID INT PRIMARY KEY,
   StudentName VARCHAR(30)
);

CREATE TABLE Faculty (
   FacultyID INT PRIMARY KEY,
   FacultyName VARCHAR(30),
   Phone VARCHAR(15)
);

CREATE TABLE Course (
   CourseID INT PRIMARY KEY,
   CourseName VARCHAR(30),
   FacultyID INT,
   FOREIGN KEY(FacultyID) REFERENCES Faculty(FacultyID)
);

CREATE TABLE Enrollment (
   StudentID INT,
   CourseID INT,
   PRIMARY KEY(StudentID, CourseID),
   FOREIGN KEY(StudentID) REFERENCES Student(StudentID),
   FOREIGN KEY(CourseID) REFERENCES Course(CourseID)
);
INSERT INTO Student VALUES
(101,'Jack'),
(102,'Hema');
```

```
INSERT INTO Faculty VALUES
(1,'Khamzat','9876543210'),
(2,'Khabib','9876501234');

INSERT INTO Course VALUES
(201,'DBMS',1),
(202,'Python',2);

INSERT INTO Enrollment VALUES
(101,201),
(102,202);
SELECT S.StudentName,C.CourseName,F.FacultyName
FROM Enrollment E
JOIN Student S ON E.StudentID=S.StudentID
JOIN Course C ON E.CourseID=C.CourseID
JOIN Faculty F ON C.FacultyID=F.FacultyID;
```

```
+-------------+------------+-------------+
| StudentName | CourseName | FacultyName |
+-------------+------------+-------------+
| Jack        | DBMS       | Khamzat     |
| Hema        | Python     | Khabib      |
+-------------+------------+-------------+
2 rows in set (0.00 sec)
```

2. A company database contains employee and department details. Design appropriate SQL queries to retrieve employees working in multiple departments and justify your approach

```
CREATE TABLE Employee(
   EmployeeID INT,
   EmployeeName VARCHAR(30),
   DepartmentID INT
);
INSERT INTO Employee VALUES
(101,'Rehan',1),
(101,'Rehan',2),
(102,'Priyansh',1),
(103,'Khabib',3),
(103,'Khabib',4);
SELECT EmployeeID,EmployeeName
FROM Employee
GROUP BY EmployeeID,EmployeeName
HAVING COUNT(DepartmentID)>1;
```

```
+------------+--------------+
| EmployeeID | EmployeeName |
+------------+--------------+
|        101 | Rehan        |
|        103 | Khabib       |
+------------+--------------+
2 rows in set (0.00 sec)
```

3. Given a relation with multiple functional dependencies, determine the candidate keys and check whether the relation satisfies BCNF. If not, decompose it.

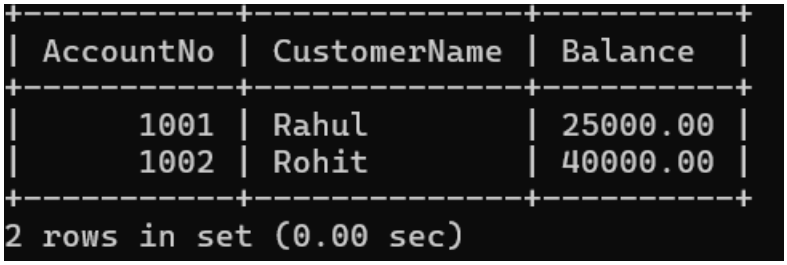
```
CREATE TABLE StudentCourse(
   StudentID INT,
   CourseID INT,
   Faculty VARCHAR(30)
);
INSERT INTO StudentCourse VALUES
(101,201,'Rehan'),
(102,202,'Saleem'),
(103,201,'Rehan');
SELECT * FROM StudentCourse;
```

```
+-----------+----------+---------+
| StudentID | CourseID | Faculty |
+-----------+----------+---------+
|       101 |      201 | Rehan   |
|       102 |      202 | Saleem  |
|       103 |      201 | Rehan   |
+-----------+----------+---------+
3 rows in set (0.00 sec)
```

4. A banking system requires secure data access. Design views and constraints to ensure data security and integrity for different types of users.

```
CREATE TABLE Account(
   AccountNo INT PRIMARY KEY,
   CustomerName VARCHAR(30),
   Balance DECIMAL(10,2)
);
INSERT INTO Account VALUES
(1001,'Rahul',25000),
(1002,'Rohit',40000);
CREATE VIEW CustomerView AS
SELECT AccountNo,CustomerName,Balance
FROM Account;
```

SELECT * FROM CustomerView;



```
+-----------+--------------+----------+
| AccountNo | CustomerName | Balance  |
+-----------+--------------+----------+
|      1001 | Rahul        | 25000.00 |
|      1002 | Rohit        | 40000.00 |
+-----------+--------------+----------+
2 rows in set (0.00 sec)
```

5. A sales database contains redundant data leading to anomalies. Analyze the schema and apply normalization up to 3NF or BCNF to eliminate redundancy.

```
CREATE TABLE Customer(
   CustomerID INT PRIMARY KEY,
   CustomerName VARCHAR(30)
);

CREATE TABLE Supplier(
   SupplierID INT PRIMARY KEY,
   SupplierName VARCHAR(30)
);

CREATE TABLE Product(
   ProductID INT PRIMARY KEY,
   ProductName VARCHAR(30),
   SupplierID INT,
   FOREIGN KEY(SupplierID) REFERENCES Supplier(SupplierID)
);

CREATE TABLE Orders(
   OrderID INT PRIMARY KEY,
   CustomerID INT,
   ProductID INT,
   FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID),
   FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
);
INSERT INTO Customer VALUES
(1,'Rohit'),
(2,'Parthiv');

INSERT INTO Supplier VALUES
(11,'indian Traders');
```

```
INSERT INTO Product VALUES
(101,'Laptop',11),
(102,'Mouse',11);

INSERT INTO Orders VALUES
(1001,1,101),
(1002,2,102);
SELECT C.CustomerName,P.ProductName,S.SupplierName
FROM Orders O
JOIN Customer C ON O.CustomerID=C.CustomerID
JOIN Product P ON O.ProductID=P.ProductID
JOIN Supplier S ON P.SupplierID=S.SupplierID;
```

```
+--------------+-------------+----------------+
| CustomerName | ProductName | SupplierName   |
+--------------+-------------+----------------+
| Rohit        | Laptop      | indian Traders |
| Parthiv      | Mouse       | indian Traders |
+--------------+-------------+----------------+
2 rows in set (0.00 sec)
```

6. Given a dataset of customers and orders, write SQL queries using joins and subqueries to retrieve meaningful information such as frequent customers.

```
CREATE TABLE Customer (
   CustomerID INT PRIMARY KEY,
   CustomerName VARCHAR(30)
);

CREATE TABLE Orders (
   OrderID INT PRIMARY KEY,
   CustomerID INT,
   FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)
);
INSERT INTO Customer VALUES
(1,'Rohit'),
(2,'Parthiv'),
(3,'Kohli');

INSERT INTO Orders VALUES
(101,1),
(102,1),
(103,1),
(104,2),
(105,2),
(106,1);
```

```
SELECT C.CustomerName, O.OrderID
FROM Customer C
JOIN Orders O
ON C.CustomerID = O.CustomerID;

SELECT CustomerID, COUNT(OrderID) AS TotalOrders
FROM Orders
GROUP BY CustomerID
HAVING COUNT(OrderID) > 3;
```

```
+--------------+---------+
| CustomerName | OrderID |
+--------------+---------+
| Rohit        |     101 |
| Rohit        |     102 |
| Rohit        |     103 |
| Rohit        |     106 |
| Parthiv      |     104 |
| Parthiv      |     105 |
+--------------+---------+
6 rows in set (0.00 sec)
```

```
+------------+-------------+
| CustomerID | TotalOrders |
+------------+-------------+
|          1 |           4 |
+------------+-------------+
1 row in set (0.00 sec)
```

7. A relation contains multi-valued dependencies. Analyze the structure and decompose it into 4NF, explaining the reasoning.

```
CREATE TABLE Student(
   StudentID INT,
   Hobby VARCHAR(30),
   Language VARCHAR(30)
);
INSERT INTO Student VALUES
(101,'Cricket','English'),
(101,'Cricket','Hindi'),
(101,'Music','English'),
(101,'Music','Hindi');
SELECT * FROM Student;
```

```
+-----------+---------+----------+
| StudentID | Hobby   | Language |
+-----------+---------+----------+
|       101 | Cricket | English  |
|       101 | Cricket | Hindi    |
|       101 | Music   | English  |
|       101 | Music   | Hindi    |
+-----------+---------+----------+
4 rows in set (0.00 sec)
```

8. A database designer used improper constraints, leading to inconsistent data. Evaluate the schema and suggest appropriate integrity constraints.

```
CREATE TABLE Employee(
   EmployeeID INT PRIMARY KEY,
   EmployeeName VARCHAR(30) NOT NULL,
   Salary DECIMAL(10,2) CHECK(Salary > 0),
   Email VARCHAR(40) UNIQUE
);
INSERT INTO Employee VALUES
(101,'Rohit',45000,'rohit@gmail.com'),
(102,'riya',42000,'riya@gmail.com');
SELECT * FROM Employee;
```

```
+------------+--------------+----------+-----------------+
| EmployeeID | EmployeeName | Salary   | Email           |
+------------+--------------+----------+-----------------+
|        101 | Rohit        | 45000.00 | rohit@gmail.com |
|        102 | Riya         | 42000.00 | riya@gmail.com  |
+------------+--------------+----------+-----------------+
2 rows in set (0.00 sec)
```

9. For a given relational schema, analyze the presence of join dependencies and decompose it into 5NF to remove redundancy.

```
CREATE TABLE SupplierProjectPart(
   Supplier VARCHAR(30),
   Project VARCHAR(30),
   Part VARCHAR(30)
);
INSERT INTO SupplierProjectPart VALUES
('Asian','P1','Groceries'),
('African','P2','Golg'),
('American','P1','Oil');
SELECT * FROM SupplierProjectPart;
```

```
+----------+---------+-----------+
| Supplier | Project | Part      |
+----------+---------+-----------+
| Asian    | P1      | Groceries |
| African  | P2      | Golg      |
| American | P1      | Oil       |
+----------+---------+-----------+
3 rows in set (0.00 sec)
```

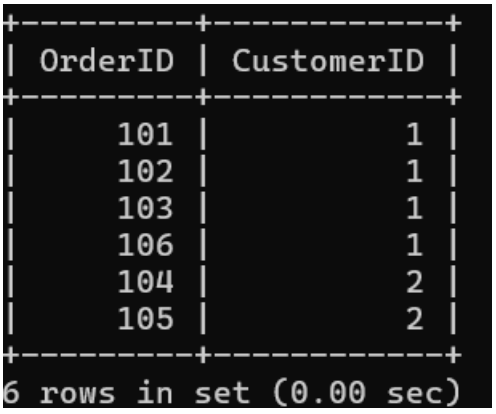
10. A system uses inefficient SQL queries for data retrieval. Analyze the queries and optimize them using joins, indexing concepts, or query restructuring.

```
CREATE TABLE Customer(
   CustomerID INT PRIMARY KEY,
   CustomerName VARCHAR(30)
);

CREATE TABLE Orders(
   OrderID INT PRIMARY KEY,
   CustomerID INT,
   FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)
);
INSERT INTO Customer VALUES
(1,'Sahul'),
(2,'riya');

INSERT INTO Orders VALUES
(101,1),
(102,2),
(103,1);

SELECT O.*
FROM Orders O
JOIN Customer C
ON O.CustomerID = C.CustomerID;
CREATE INDEX idx_customer
ON Orders(CustomerID);
```



```
+---------+------------+
| OrderID | CustomerID |
+---------+------------+
|     101 |          1 |
|     102 |          1 |
|     103 |          1 |
|     106 |          1 |
|     104 |          2 |
|     105 |          2 |
+---------+------------+
6 rows in set (0.00 sec)
```